Fixing DataCo's late deliveries, lost margin and fraud

Supply chain analysis · 65,752 orders · Jan 2015 – Jan 2018

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Data Analyst · SQL · Python · Power BI

Three fixes could add $1.14M profit and cut late deliveries from 57% to 34%

LATE DELIVERIES

57.3%

of shipped orders arrive late

34.3%

Fix: promise delivery times we can actually meet

LOSS-MAKING DISCOUNTS

−$3.88M

lost on items sold below cost

+$1.14M

Fix: cap discounts at 10%

PAYMENT FRAUD

$741K

of suspected fraud, 18.7% of profit

100%

Fix: hold bank transfers until paid, catching all of it

180,519 order lines were cleaned and modelled before any analysis

1

Raw data

180,519 order lines, 53 columns, Jan 2015 – Jan 2018

2

SQL cleaning

Fixed types, rounding and duplicates; every change logged

3

Python analysis

Explored delivery, margin and fraud patterns in pandas

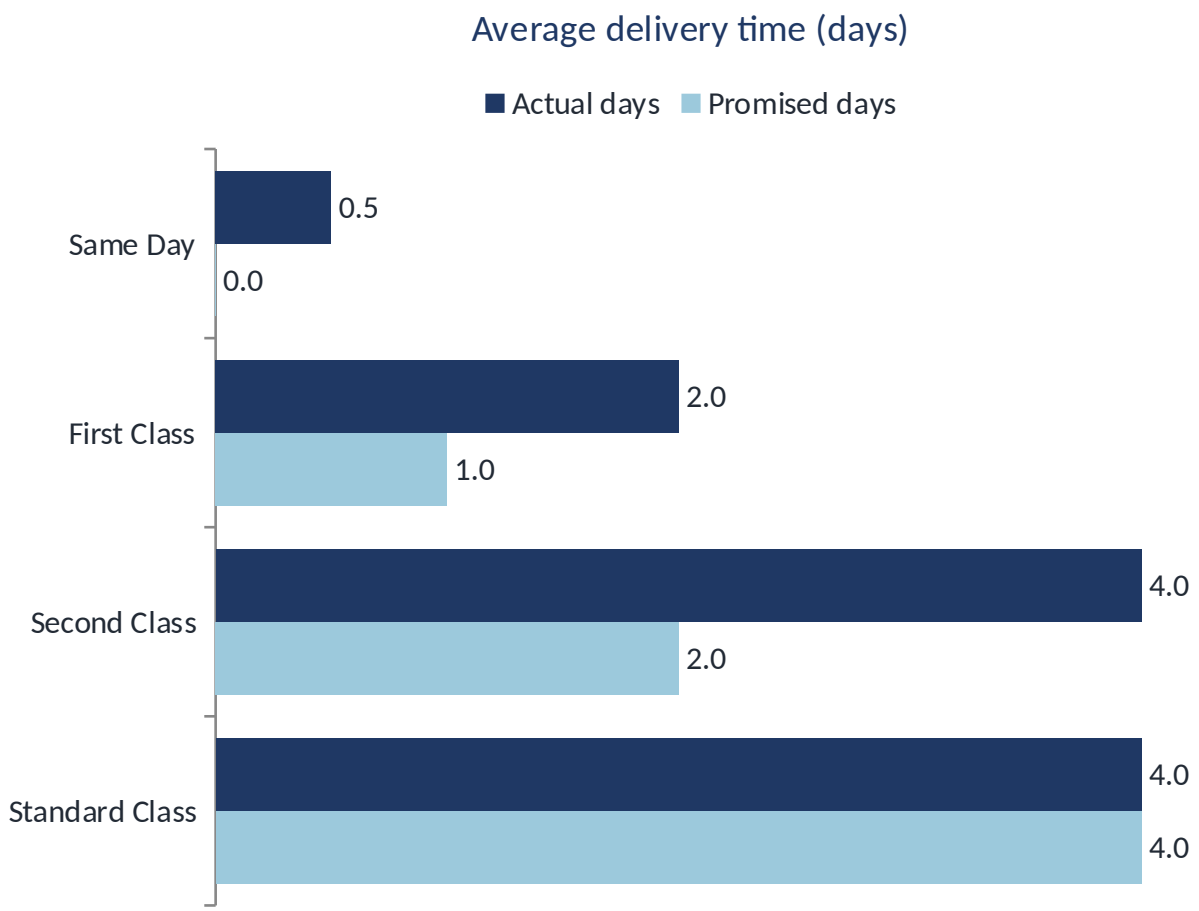
4

Power BI model

Galaxy schema: 2 fact tables, shared dimensions, 70+ DAX measures

Every cleaning decision is recorded in the data quality log, so each number in this deck can be traced back to the raw data.

57% of orders are late because we promise faster than we deliver



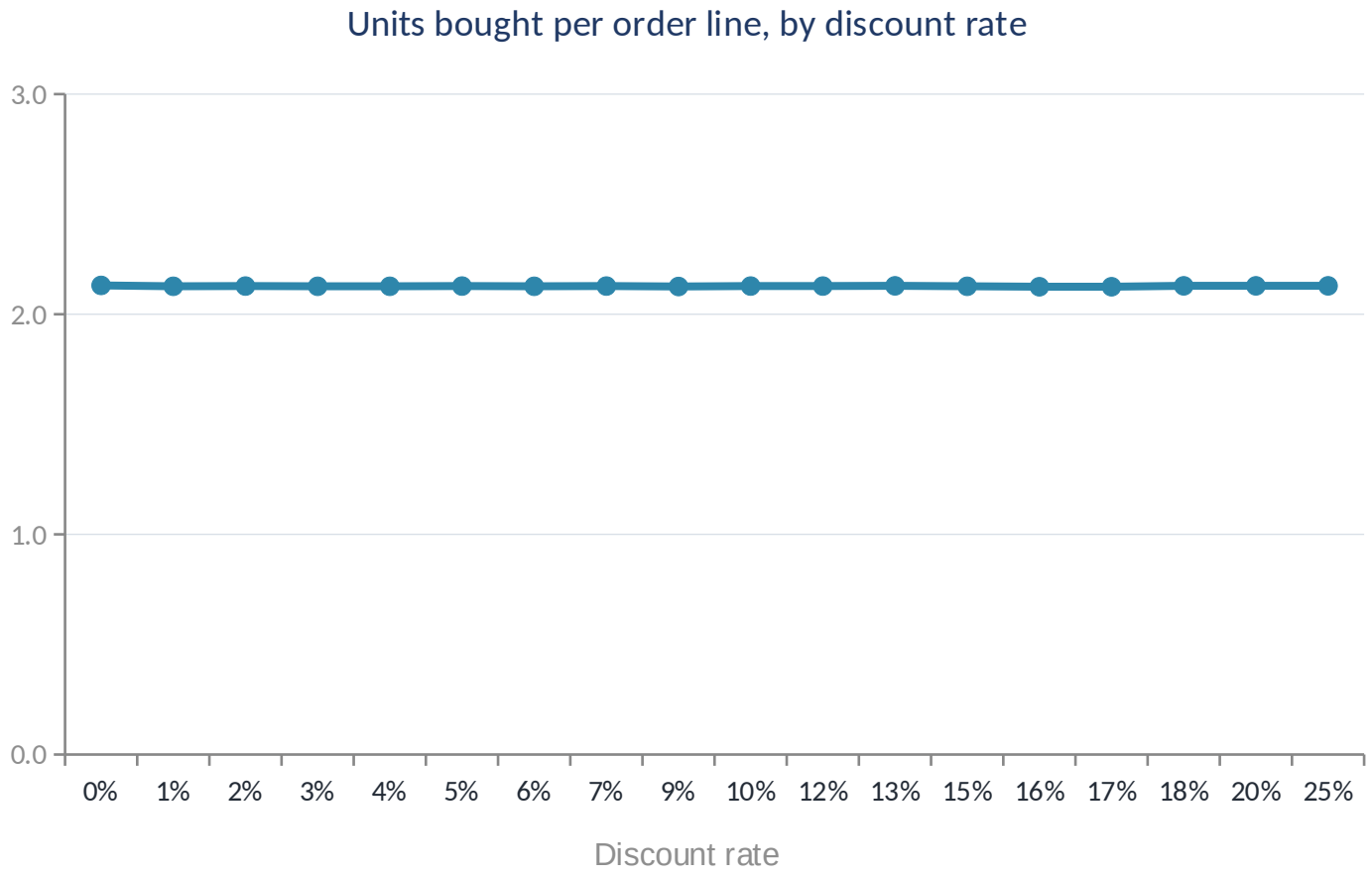
100%

of First Class orders arrive late, in every market

34.3%

late rate if First Class promises 2 days and Second Class 4 (from 57.3%)

Bigger discounts don't sell more, they just give margin away



18.7%

of items sell at a loss, costing $3.88M over three years

+$1.14M

extra profit from a 10% discount cap (+28.9%, about $367K in 2017)

Every suspected fraud order was paid by bank transfer

Share of orders flagged as suspected fraud, by payment type



100%

of suspected fraud caught

27.5%

of orders held for checks

8.2%

of held orders are fraud

112

checks a week (~3 an hour)

Rule tested: hold bank-transfer orders until payment clears

Start with the delivery promise: it costs nothing and works immediately

	Action	Expected impact	Effort
1	**Reset delivery promises: First Class 2 days, Second Class 4 days**	Late rate falls from 57.3% to 34.3%	Low: a policy change, no extra cost
2	**Hold bank-transfer orders until payment clears**	Protects about $256K of profit a year	Medium: about 112 checks a week
3	**Cap discounts at 10%, then review loss-making lines**	About +$367K profit a year (2017 basis)	Low: monitor volume for 1–2 quarters

Explore the full interactive dashboard

Overview

Revenue, margin, late rate and fraud at a glance

Delivery

Promised vs actual days and late-rate heatmap

Profitability

Discount analysis with a what-if slider

Fraud

Payment-type risk and the hold-rule test

Code, SQL, notebooks and Power BI file
github.com/GLICON/dataco-supply-chain-analysis

Notes: fraud means orders labelled suspected fraud, not confirmed cases. delivery_status was excluded from the fraud analysis because it is set after the fraud decision (data leakage). Data covers Jan 2015 – Jan 2018 from the public DataCo dataset; DataCo is not a real client.